



Routes Manipulation & NAT

ACE Team

RFC1918 Routes Injection = Standard behavior of the Controller

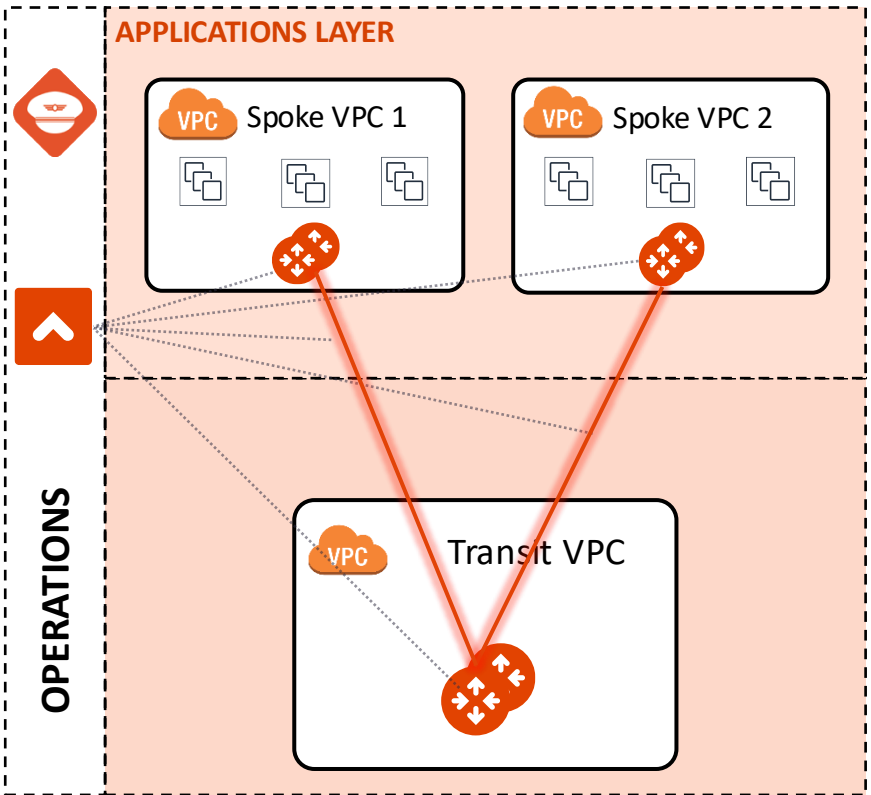
- Once the Controller completes deploying the **attachments** between the Spoke Gateways and Transit Gateways, it will configure the three **RFC 1918** routes in the route tables to point to the ENIs of the Spoke Gateways.

Private Route table

Routes	Subnet associations	Edge associations	Route propagation	Tags
Routes (4)				
Filter routes				
Destination			Target	
10.0.0.0/8			eni-08ac50fc16cd8c4a5	
10.1.1.0/24			local	
172.16.0.0/12			eni-08ac50fc16cd8c4a5	
192.168.0.0/16			eni-08ac50fc16cd8c4a5	

Public Route table

Routes	Subnet associations	Edge associations	Route propagation	Tags
Routes (5)				
Filter routes				
Destination			Target	
0.0.0.0/0			igw-07c6ddedd190d12d3	
10.0.0.0/8			eni-08ac50fc16cd8c4a5	
10.1.1.0/24			local	
172.16.0.0/12			eni-08ac50fc16cd8c4a5	
192.168.0.0/16			eni-08ac50fc16cd8c4a5	

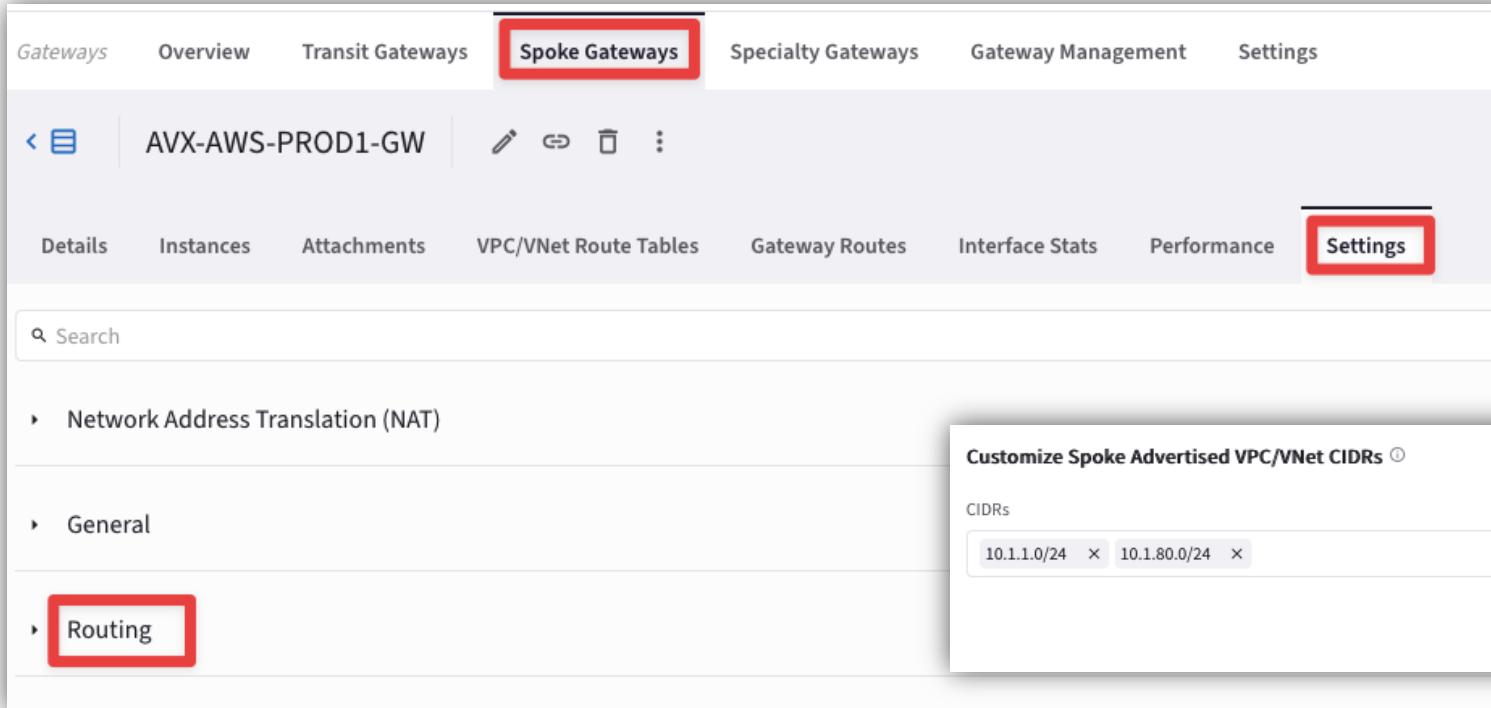


Routes Manipulation – Customize Spoke Advertised VPC CIDRs

- For example, if you decide to add another CIDR block to an existing VPC, in that case you can also advertise the new CIDR within your CNSF.

CIDRs		Flow logs	Tags
CIDRs		Info	
Address type		CIDR	
IPv4		10.1.1.0/24	
IPv4		10.1.80.0/24	

- PATH:** COPILOT > Cloud Fabric > Gateways > Spoke Gateways > select the relevant GW > Settings > Routing > **Customize Spoke Advertised VPC/VNet CIDRs**



Gateways Overview Transit Gateways **Spoke Gateways** Specialty Gateways Gateway Management Settings

< AVX-AWS-PROD1-GW

Details Instances Attachments VPC/VNet Route Tables Gateway Routes Interface Stats Performance **Settings**

Search

Network Address Translation (NAT)

General

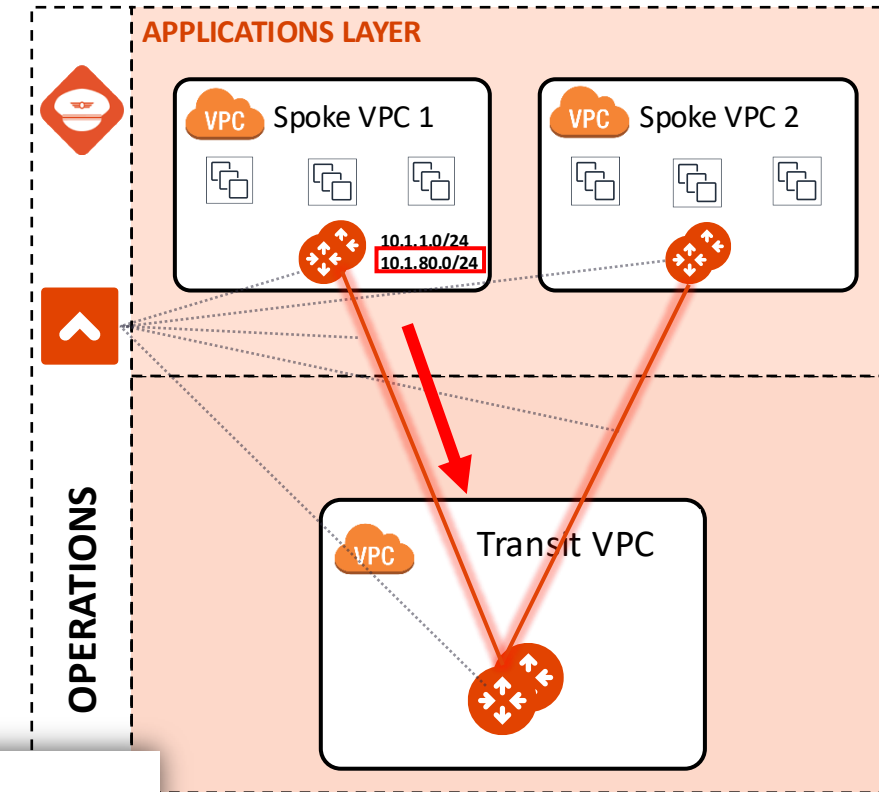
Routing

Customize Spoke Advertised VPC/VNet CIDRs

CIDRs

10.1.1.0/24 × 10.1.80.0/24 ×

Cancel Save



Routes Manipulation – Customize Spoke VPC Routing Table

- You can also decide to withdraw an RFC1918 route that was previously injected by the Controller, customizing the VPC/VNet/VCN routing table.

Filter routes	
Destination	Target
10.0.0.0/8	eni-019dbf18ad5b8857f
10.1.3.0/24	local
172.16.0.0/12	eni-019dbf18ad5b8857f
192.168.0.0/16	eni-019dbf18ad5b8857f

- PATH:** COPILOT > Cloud Fabric > Gateways > Spoke Gateways > select the relevant GW > Settings > Routing > **Customize Spoke VPC/VNet Route Table**

Gateways Overview Transit Gateways **Spoke Gateways** Specialty Gateways Gateway Management Settings

AVX-AWS-PROD1-GW

Details Instances Attachments VPC/VNet Route Tables Gateway Routes Interface Stats Performance **Settings**

Search

Network Address Translation (NAT)

General

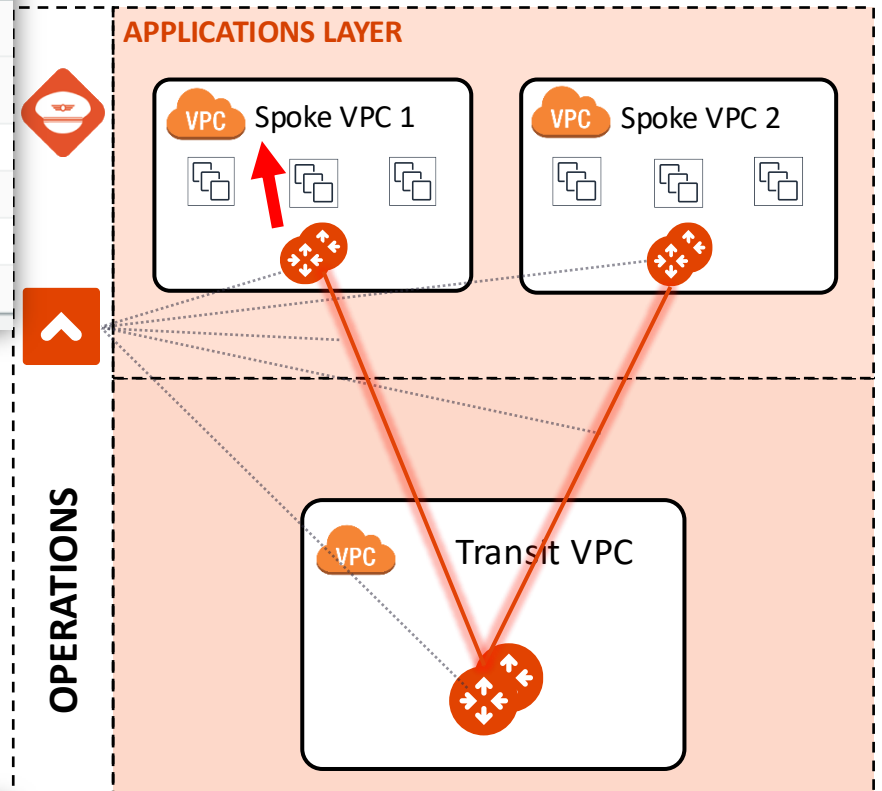
Routing

Customize Spoke VPC/VNet Route Table

CIDRs

172.16.0.0/12 x 192.168.0.0/16 x

Cancel Save



Routes Manipulation – Exclude Learned CIDRs to Spoke VPC/VNet Route Table

Gateway Instance

AVX-AWS-SPOKE-GW-PROD1

Network Domain

All

▼	Destination	Via	Interface	Next Hop IP	Next Hop Gateway	Metric
	10.1.1.0/24	10.1.1.49	eth0	10.1.1.49		0
	10.1.1.49/32		eth0			100
	10.1.2.0/24 , + 2 more		tun-0A010147-0	10.1.1.71	AVX-AWS-SPOKE-GW-PROD...	200

You can also decide to delete a more specific route from the routing table of a Spoke GW.

- PATH:** COPILOT > Cloud Fabric > Gateways > Spoke Gateways > select the relevant GW > Settings > Routing > **Exclude Learned CIDRs to Spoke VPC/Vnet Route Table**

Gateways

Overview

Transit Gateways

Spoke Gateways

Specialty Gateways

Gateway Management

Settings

< ☰

AVX-AWS-PROD1-GW

✎ 🔗 🗑️ ⋮

Details

Instances

Attachments

VPC/VNet Route Tables

Gateway Routes

Interface Stats

Performance

Settings

🔍 Search

▸ Network Address Translation (NAT)

▸ General

▸ Routing

Exclude Learned CIDRs to Spoke VPC/VNet Route Table ⓘ

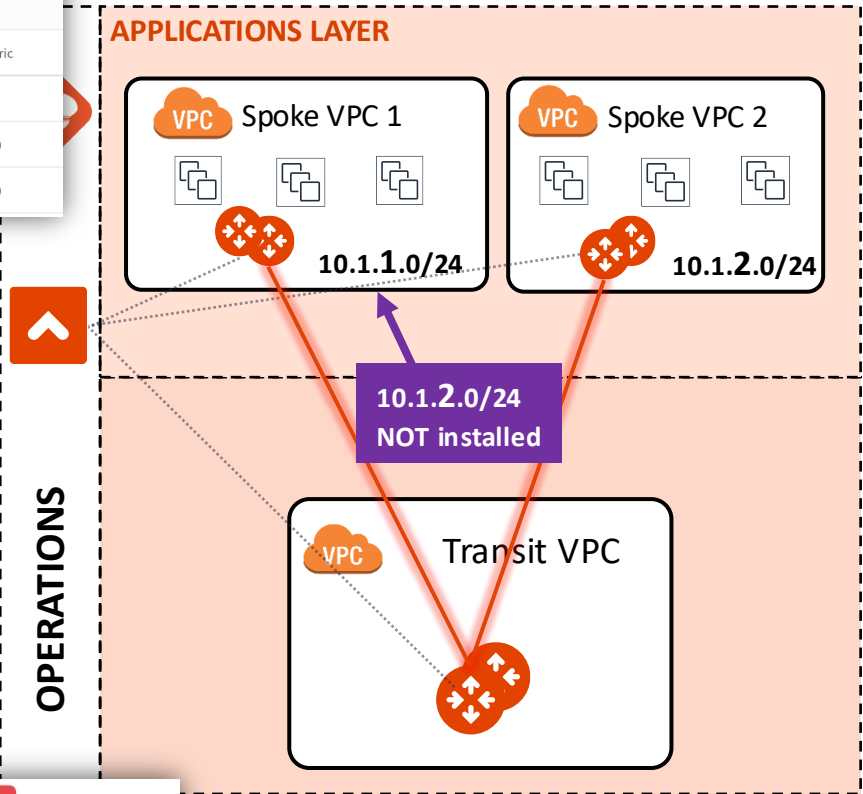
CIDRs

10.1.2.0/24 ×

×

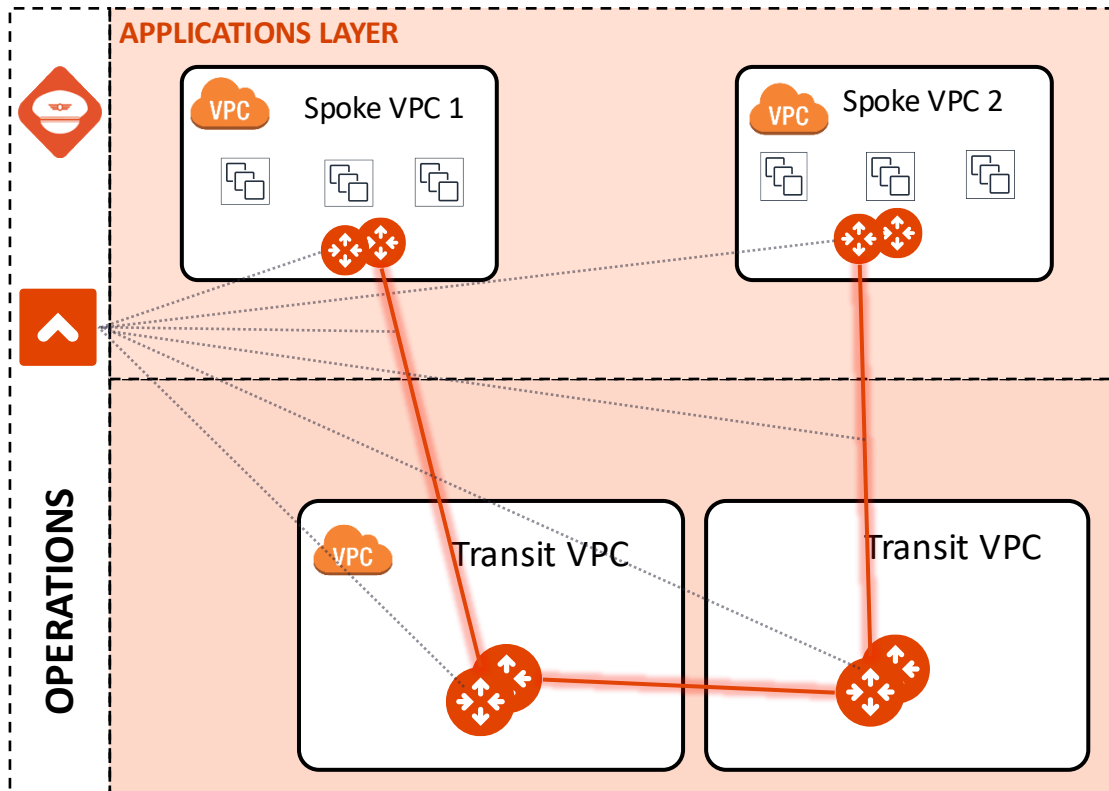
Cancel

Save

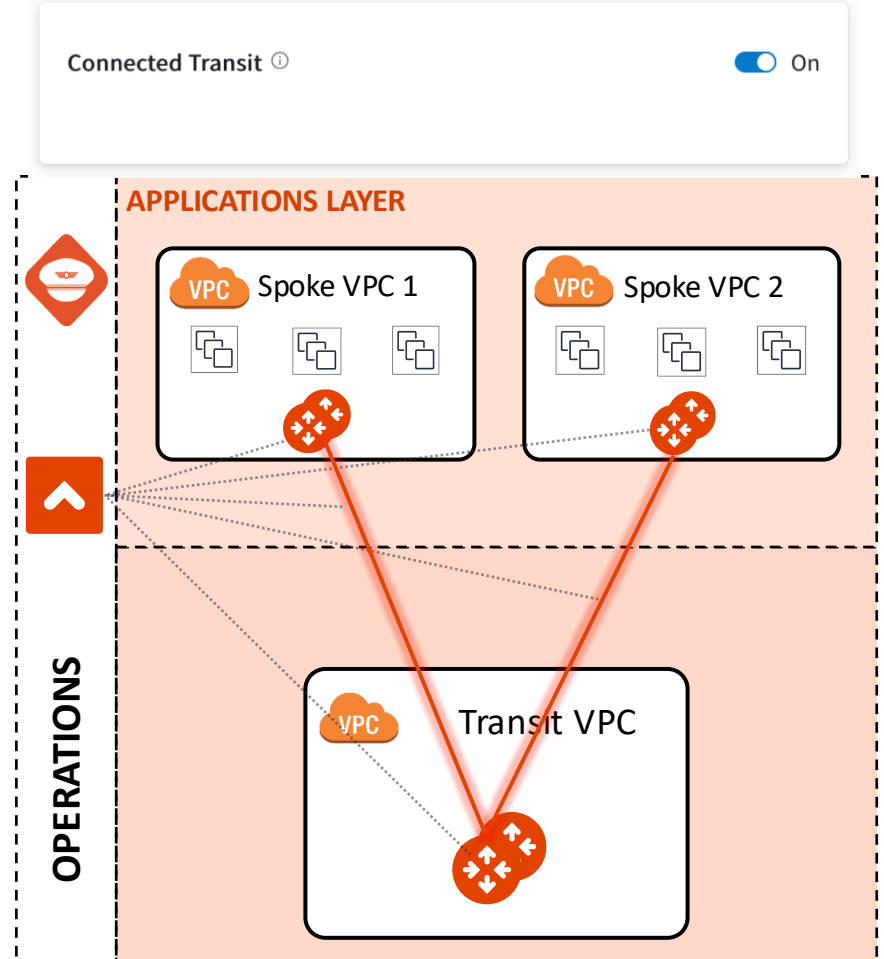


Routes Manipulation – Connected Transit

- **By default**, Aviatrix Spoke VPCs/VNets do not have routing established to communicate with each other via Transit. They are completely segmented.
- Each Spoke VPC should be connected to its own dedicated Transit VPC.



- If you would like to build a full mesh network where Spoke VPCs/VNets communicate with each other via Transit GW, you can achieve that by enabling **Connected Transit** mode.



NAT - Overview

- The Aviatrix Spoke Gateways provide advanced NAT capabilities.
- Three NAT functions are supported:
 - ❑ **Source NAT:**
 - Two modes of Source NAT are supported:
 - Single IP
 - Customized SNAT
 - ❑ **Destination NAT**
 - ❑ **Mapped NAT**

Source NAT: Single IP



Gateways Overview Transit Gateways **Spoke Gateways**

< AVX-AWS-SPOKE-GW-PROD1

Instances Connections VPC/VNet Route Tables Gateway Routes Interfaces Route DB **Settings**

Search

Network Address Translation (NAT)

Source NAT ⓘ

☒ Single IP ☐ Customized SNAT

On

Cancel Save

- **PATH:** COPILOT > Cloud Fabric > Gateways > Spoke Gateways > select the relevant gateway > Settings > **NAT**
- In addition to the RFC1918 routes, the Controller will also install a *quad zero* route that points to the ENI of the Spoke Gateway.

- ^ Network Address Translation (NAT)

Source NAT ⓘ

Single IP

Customized SNAT

Instance

AVX-AWS-SPOKE-GW-PROD1

+ Rule

Filter

Grid

Download

Unsaved Changes: Add: 1

Search

Src CIDR	Src Port	Dst CIDR	Dst Port	Protocol	Connection	Mark	SNAT IPs	SNAT Port	Apply Route Entry	Exclude Route Table	
10.1.1.0/24		10.1.2.0/24		all	None		192.168.1.0/24				

Total 1 Rule

Cancel

Save

Destination NAT

- Destination NAT (**DNAT**) allow you to change the destination to a virtual address range.

^ Network Address Translation (NAT)

Source NAT ⓘ Off

Destination NAT ⓘ On

Instance
ace-aws-eu-west-1-spoke1

[+ Rule](#) | [Filter](#) [Grid](#) [Download](#) | Unsaved Changes: [Add: 1](#) |

Src CIDR	Src Port	Dst CIDR	Dst Port	Protocol	Connection	Mark	DNAT IPs	DNAT Port	Apply Route Entry	Exclude Route Table	
10.1.1.0/24		172.16.211.0/24		all	None		50.50.50.1-50.50.50.254		Off		

Total 1 Rule ↺

[Cancel](#) [Save](#)



Tools for Operating your Routes Manipulation & NAT

Routes Manipulation - ACTIONS

- **PATH:** COPILOT > Cloud Fabric > Gateways > Spoke Gateways > select the relevant GW > Settings > **Routing**

^ Routing

Configure Private VPC/VNet Default Route ⓘ

Off

Skip Public VPC/VNet Route Table ⓘ

Off

Auto Advertise Spoke Site2Cloud CIDRs ⓘ

Off

Customize Spoke VPC/VNet Route Table ⓘ

CIDRs

Exclude Learned CIDRs to Spoke VPC/VNet Route Table ⓘ

CIDRs

Customize Spoke Advertised VPC/VNet CIDRs ⓘ

CIDRs

Update Encrypted Spoke VPC/VNet CIDRs

Update

NAT



- **PATH:** COPILOT > Fabric > Gateways > Spoke Gateways > select the relevant GW > Settings > **NAT**

^ Network Address Translation (NAT)

Source NAT ⓘ

Single IP

Customized SNAT

On

Cancel

Save

Destination NAT ⓘ

On

Instance

AVX-AWS-SPOKE-GW-PROD1

+ Rule

Search

Src CIDR	Src Port	Dst CIDR	Dst Port	Protocol	Connection	Mark	DNAT IPs	DNAT Port	Apply Route Entry	Exclude Route Table
No Rules										

Total 0 Rules



Next:

Lab 5 Routes Manipulation & NAT